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Beyond pipelines: Multi-modal CO_2 transport enables timely CO_2 capture, transport, and storage deployment

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ABSTRACT

CO_2 capture, transport, and storage (CCTS) is essential to meet climate targets, but its success is threatened by the slow expansion of pipelines, due to regulatory hurdles and public opposition. The resulting deployment limits are insufficiently accounted for in current infrastructure optimization models, which therefore overestimate the speed of expanding CCTS value chains. Here, we expand the energy system optimization model ZEN-garden to consider temporal diffusion limits for pipelines and analyze the consequences of constrained deployment for a case study of Switzerland. First, we assess the risk of missing climate targets due to the constrained deployment of CO_2 pipelines, and second, we evaluate strategies to mitigate this risk. We find that realistic pipeline diffusion speeds lead to a 9 % deficit in CO_2 sequestration compared to climate targets. The additional deployment of non-pipeline transport modes circumvents these limits and enables compliance with policy goals at a cost increase of 5 %. Such a multi-modal transport system proves substantially more cost-effective than compensating the CO_2 sequestration gap via carbon removal. Based on our findings, multi-modal transport should play an integral role in the design of CO_2 transport infrastructure to mitigate the risk posed by delays and slow pipeline deployment. The results suggest that an exclusive focus on pipelines for CO_2 transport may jeopardize the success of the entire CO_2 value chain.

1. Introduction

CO_2 capture, transport, and storage (CCTS) is a central element in many national strategies to decarbonize the economy, particularly for industrial point sources (IPCC, 2022; BFE, 2021; Levina et al., 2023). Emitters, such as cement plants, waste incinerators, or steel producers, aim to capture CO_2 to avoid purchasing CO_2 certificates from Emission Trading Systems (ETS) (Levina et al., 2023), to offer products with a green premium (Heidelberg Materials, 2025), or to receive tax credits for stored CO_2 (Global CCS Institute, 2025). Globally, the 1.5°C goal requires more than 7 Gt_{CO_2} to be captured by 2050 (IEA, 2021), a more than 100-fold increase compared to today's 64 Mt capacity (Global CCS Institute, 2025). While the number of announced CCTS projects is increasing annually, operational CO_2 capture capacity remains insufficient to fulfill climate agreements (Global CCS Institute, 2025; Kazlou et al., 2024). As no mature CO_2 transport infrastructure is available in most regions, deployment rates will depend on the number of first-of-a-kind projects planned, their success rate, and a high-growth phase in the continued deployment of the respective technology (Kazlou et al., 2024). In the past, energy infrastructure was expanded rapidly in response to energy or pollution crises, which motivated the investments. Even if CCTS technologies grow at a similar rate, they will likely fall short of international climate targets and threaten successful decarbonization (Kazlou et al., 2024). Hence, in regions without existing CO_2 infrastructure, the achievability of CO_2 sequestration goals is unclear.

In Europe, the first CCTS project dedicated to industrial decarbonization became operational in 2025. Yet, climate targets depend on a rapid, large-scale deployment of CCTS infrastructure, reaching dozens of megatonnes of captured CO_2 annually by 2050 (European Commission, 2024; BFE, 2021; European Commission et al., 2024). Many industrial emitter sites in Europe are located far from the North Sea (d'Amore et al., 2021a), where most CO_2 storage potential and the most developed injection sites are located (Clean Air Task Force, 2024c). Therefore, CO_2 must be transported from the emitter to the storage location. This step has proven challenging, especially for early European projects that cannot benefit from an existing transport infrastructure (Oeuvray et al., 2024; Becattini et al., 2024). Scaling up CO_2 transport is, therefore, a major hurdle for industries located far from storage sites (ZEP, 2020).

CO_2 can be transported via a variety of on- and offshore transport modes: Onshore modes include trains, trucks, barges, and pipelines (ZEP, 2020). While trains, trucks, and barges offer low upfront costs and high operational flexibility, they also present logistical challenges:

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switching between vehicles requires dedicated transfer terminals and buffer storage (Danish Energy Agency, 2024; ZEP, 2020). For large transport volumes, the number of additional vehicles becomes excessive and may congest roads, railway corridors, and harbor infrastructure (Oeuvray et al., 2024). Onshore pipelines, on the other hand, provide the lowest specific transport cost for large volumes (Oeuvray et al., 2024; ZEP, 2025). Additionally, onshore pipelines have the lowest environmental impacts per tonne of CO_2 transported (Nöhl et al., 2025). Offshore transport of CO_2 is carried out by ships and pipelines. Similar to onshore transport, pipelines are more cost-effective for large-scale operations (ZEP, 2020). Ships are cost-competitive with offshore pipelines only over long distances and for quantities up to several Mt_{CO_2} per year (Oeuvray et al., 2024; Danish Energy Agency, 2024).

To investigate the design and time-dependent expansion of CO_2 transport, energy system models based on mathematical optimization are commonly used in the literature. Although the models differ in geographical and sectoral scope, most studies focus on pipeline transport because of its economic benefits. For the European scope, Morbee et al. (2012) projected a pipeline network of several thousand kilometers for the year 2025. Although the timing shifted, the need for a European transport network, and its timely deployment, are well established in the literature (d'Amore and Bezzo, 2017; Sievert et al., 2025). Similar pipeline-based networks emerge when considering uncertainty in storage (Stewart et al., 2014; d'Amore et al., 2019), adding maritime ships to the set of CO_2 transport modes (d'Amore et al., 2021b), and capturing emissions from the industrial and power sectors (d'Amore et al., 2021a) as well as including bio-energy carbon capture (Negri et al., 2021). Even for a geographical scope limited to CO_2 sources from a single country, on- and offshore pipelines are the preferred transport mode (Bennæs et al., 2024). Studies that include trucks, trains, and barges as transport modes conclude that these modes can serve as interim solutions until transport volumes are large enough to warrant pipelines (Becattini et al., 2022).

While cost, emissions, and operational logistics favor pipelines (Corsten et al., 2013; Stewart et al., 2014; Danish Energy Agency, 2024), their construction is the most complex and time-consuming among all transport modes (Yeates et al., 2024; Douglas, 2025). Constructing new CO_2 pipelines requires reaching agreements with private and public stakeholders to obtain right-of-way and provide financial compensation (Peletiri et al., 2018). In addition to ownership issues, communities along the pipeline route may object to its routing or the entire construction (Gough et al., 2014). Such opposition may prevent or delay new transport projects by several years, even in countries where CO_2 pipelines already exist (George, 2023). These challenges lead to considerably longer construction times for pipeline projects than for maritime ships (Noothout et al., 2014; Northern Lights JV DA, 2024b). In addition to its long construction time, large-scale pipeline infrastructure requires new industrial capabilities, as Europe currently lacks experience in constructing and operating CO_2 pipelines. The speed at which companies can develop such capabilities also affects the timeline of the entire network.

Nevertheless, studies rarely assess the challenges associated with building pipelines and the feasibility of scaling up the pipeline network (Morbee et al., 2012; Bennæs et al., 2024; Becattini et al., 2022). Therefore, the expansion of the transport network is not perceived as a hurdle (Martin-Roberts et al., 2021). When considering the deployment speed of CO_2 technologies, scholars apply the limits only to CO_2 capture units of specific industrial emitters but not to the transport infrastructure (Wei et al., 2021). However, scaling up non-existent technologies is challenging and should be explicitly considered in transition pathway models (Leibowicz et al., 2016). In publications focusing on the general energy system deployment, scholars already apply deployment limits to CCTS technologies (Mannhardt et al., 2024). Such studies examine the broader energy transition but do not address the rollout of CCTS transport infrastructure in detail. Other studies conclude that the required expansion speed derived from modeling results is not unprecedented and is similar to the historical expansion rate of the natural gas pipeline industry (Neele et al., 2011). However, equating CO_2 and natural gas pipelines ignores essential differences between them: On the one hand, CO_2 pipelines face greater opposition, which causes delays or outright project cancellations (George, 2023, 2025; Douglas, 2025). On the other hand, CO_2 pipelines lack regulation in Europe, creating uncertainty about how they can be constructed and operated, especially for cross-border transport (Frattini et al., 2024; European Commission, 2025). Besides legislation, the large number of canceled and retracted projects continues to threaten the expansion of CO_2 transport infrastructure. Among all announced CCTS projects, fewer than half reach the operational stage (Martin-Roberts et al., 2021; Abdulla et al., 2021). Cancellations of CCTS projects are more likely when projects have high capital costs and lack a clear revenue stream from subsidies or commercial interests. CO_2 pipelines are inherently capital-intensive and often lack a clear business case with consistent revenue, with the exception of enhanced oil recovery (Abdulla et al., 2021). Such a combination suggests a higher failure rate and, thus, a slower overall pace of CO_2 transport deployment.

In other studies, the authors exogenously impose a constant limit on pipeline deployment (d'Amore and Bezzo, 2017). However, the capacity to deploy new technologies is not constant over time and instead follows an S-shaped curve (Kazlou et al., 2024). Incorporating an S-shaped curve into an optimization model requires endogenous consideration of deployment limits. An endogenous approach is currently missing from the transition pathway literature on CCTS systems, since previous literature often treats expansion as a constant. Therefore, there is no evidence in the literature on whether the pipeline deployment speed will become a bottleneck for the timely construction of a large-scale CCTS system in Europe.

Here, we investigate the scale-up of European CO_2 transport infrastructure to determine whether, and under what circumstances, pipeline deployment limits threaten climate targets. To this end, our contribution is threefold: First, we integrate temporal technology diffusion limits, based on Leibowicz et al. (2016), into a transition pathway model (Mannhardt et al., 2025) and test their applicability to CO_2 pipelines. Second, we optimize and assess the expansion of Swiss CCTS infrastructure through 2050, subject to the diffusion limits, using scenarios. The consequences of slow pipeline deployment on climate targets are quantified. Third, the model identifies solutions to mitigate the consequences of the slow deployment. We extend the current perspective on diffusion by introducing new ways to model these limits, accounting for the specific challenges of pipeline construction. Whereas previous models base diffusion solely on a technology's production or transport capacity, we also include the transport distance. Using the distance, the diffusion limit captures the challenge of constructing a pipeline across large regions. Based on a case study of the Swiss industrial sector, we explore feasible options for successfully scaling up the CCTS infrastructure. We show that using multi-modal transport, where pipelines are complemented by alternatives such as barges and ships, enables expanding the CO_2 transport network despite deployment limitations. Establishing local CO_2 storage provides similar benefits but is more uncertain and has longer lead times than non-pipeline transport modes. Thus, while the limited pipeline deployment can be a bottleneck in reaching CCTS targets, this risk can be mitigated with existing technologies.

The remainder of the article is structured as follows: Section 2 outlines the optimization model and the integration of deployment limits, followed by a description of the case study. Section 4 presents the case study results and their interpretation across all scenarios. Section 5 discusses the implications of the results and concludes the article.

2. Methods

2.1. Modeling framework

We minimize the net present cost (Eq. (1)) of expanding and operating the CCTS infrastructure using the energy system optimization tool *ZEN-garden*. The tool can model the spatial and temporal deployment of technologies within a system transition. In the following, we give a brief overview of the cost calculations and emissions accounting. We refer to the ZEN-garden documentation for details on mass balance, operational, and other constraints (Mannhardt et al., 2025; Reliability and Risk Engineering lab, 2025).

$$\min \sum_{y \in \mathcal{Y}} DF_y \left(CAPEX_y^{ann} + OPEX_y^{tech} + OPEX_y^{carrier} + c^{pen} \cdot E_y^{pen} \right) \quad (1)$$

All future annualized investment ($CAPEX_y^{ann}$) and operating ($OPEX_y$) cash flows are discounted with the discount factor DF_y to the initial planning period y_0 in the optimization horizon $\mathcal{Y}$. The investment costs are annualized over the technologies' lifetime (Reliability and Risk Engineering lab, 2025). The optimization horizon $\mathcal{Y}$ comprises the planning periods y which have an interval of dy years between them. The interval is considered in computing the discount factor (Eq. (2)). The term $c^{pen} \cdot E_y^{pen}$ represents a penalty cost for missing emission goals, such as annual emission limits, emission budgets, or CO_2 sequestration goals. The penalty cost c^{pen} is set high enough such that penalty emissions E_y^{pen} are only produced by the optimization to avoid infeasible solutions.

$$DF_y = \begin{cases} \sum_{i \in [0, dy-1]} \left(\frac{1}{1+r} \right)^{dy(y-y_0)+i} & \text{if } y < \max(\mathcal{Y}), \\ \left(\frac{1}{1+r} \right)^{dy(\max(\mathcal{Y})-y_0)} & \text{otherwise.} \end{cases} \quad (2)$$

Main outputs of the transition pathway optimization are the total built capacities $S_{h,p,y}$ of the technologies h. For any given location p and planning period y, the total capacity $S_{h,p,y}$ of technology h is the sum of previous capacity additions $\Delta S_{h,p,y}$ that have not yet reached their lifetime l_h. Both the total capacity $S_{h,p,y}$ and the capacity additions $\Delta S_{h,p,y}$ are limited to $s_{h,p,y}^{max}$ and $\Delta s_{h,p,y}^{max}$, respectively:

$$S_{h,p,y} = \sum_{\bar{y}=\max\left(y_0, y-\left\lfloor \frac{l_h}{dy} \right\rfloor+1\right)}^{y} \Delta S_{h,p,\bar{y}} \quad (3)$$

$$0 \leq S_{h,p,y} \leq s_{h,p,y}^{\max} \quad (4)$$

$$0 \leq \Delta S_{h,p,y} \leq \Delta s_{h,p,y}^{\max} \quad (5)$$

In this study, capacity limits are required for non-pipeline transport modes, as size limitations are a disadvantage compared to pipelines. The driver for building the CCTS infrastructure in our model is a minimum target for sequestering CO_2. Sequestered CO_2 is often a direct component of countries' climate strategies, and can be directly implemented without additional assumptions, e.g., on how to allocate emission budgets:

$$E_y^{pen} + \sum_n \overline{U}_{CO_2-stored,n,y} \geq e_y^{CO_2-stored}. \quad (6)$$

The minimum sequestration target sets a lower bound for storing CO_2 ($e_y^{CO_2-stored}$) for each planning period y. The model enforces the lower bound at the storage sites, where the stored CO_2 leaves the system boundary. The amount of CO_2 exiting the system $\overline{U}_{CO_2-stored,n,y}$ is summed over all nodes n. The approach accounts for direct CO_2 losses within the supply chain, as these reduce the total amount reaching the storage sites. The model includes life-cycle emission factors for energy inputs and technology installation and operation (Burger et al., 2024; Nöhl et al., 2025). However, life-cycle emissions of the CCTS chains are excluded from the CO_2 balance in Eq. (6). Policy goals generally only include direct emissions in their sequestration targets, which is well approximated by our approach. The sequestration target includes the penalty variable E_y^{pen} for missed emission targets (see Eqs. (1) and (6)). The variable allows quantifying the gap between feasible CO_2 sequestration and policy goals when infrastructure is constrained.

The modeled CCTS system comprises three distinct parts: capture, transport, and storage of CO_2. On the capture side, the emitters produce a flue gas which, when leaving the system, causes greenhouse gas emissions. To abate industrial emissions, the CO_2 can be separated from the flue gas via capture units. Compression or liquefaction prepares the CO_2 for transport, depending on the requirements of the subsequent transport mode. We assume identical impurity requirements for all transport modes and, therefore, do not explicitly model the purification step. The transport part consists of a range of onshore and offshore transport modes: onshore connections based on trucks, trains, and barges require liquefied CO_2, whereas onshore pipelines use compressed supercritical CO_2. For offshore transport, two transport modes are included: offshore pipelines, which transport supercritical CO_2, and maritime ships, which require liquid CO_2 (Oeuvray et al., 2024; IPCC, 2005).

Unitary pipeline costs decrease with increasing transport capacity, as pipeline construction costs depend primarily on distance (Oeuvray et al., 2024; IPCC, 2005). The dependence on distance leads to poor representation of the resulting expenses when capital expenditures (CAPEX) are linearly scaled with capacity, as is common in linear energy system models (Bröchin et al., 2024). A more accurate representation requires the introduction of binary variables to approximate the non-linear behavior of CAPEX, following standard practice in CCTS literature (d'Amore and Bezzo, 2017; d'Amore et al., 2019; Bennæs et al., 2024; Middleton et al., 2020). We follow this approach and use a mixed integer linear program to optimize the transition pathway. The binary variables $g_{j,e,y}$ represent the decision to install a pipeline on an edge $e \in \mathcal{E}$ in investment period $y \in \mathcal{Y}$. The CAPEX of on- and offshore pipelines is calculated as:

$$CAPEX_{j,e,y} = g_{j,e,y} \cdot d_{j,e} \cdot CAPEX_j^{distance} + \Delta S_{j,e,y} \cdot CAPEX_j^{capacity} \quad \forall j \in \{\text{onshore pipeline, offshore pipeline}\}. \quad (7)$$

Unlike the other transport technologies, the CAPEX of pipelines depends on both the capacity addition $\Delta S_{j,e,y}$ and the distance $d_{j,e}$, with parameters $CAPEX_j^{capacity}$ and $CAPEX_j^{distance}$, respectively (Becattini et al., 2022). The result is a linear investment cost curve with an offset due to $g_{j,e,y} \cdot d_{j,e} \cdot CAPEX_j^{distance}$, approximating the economies of scale while keeping the computational complexity low.

The last part of the CCTS chain is the permanent geological CO_2 storage, which is modeled as a CO_2 sink at the storage sites. In the model, CO_2 capture, transport, and permanent storage at a sink prevents the release of industrial flue gases and, thus, reduces overall emissions. Due to large uncertainty in the cost data and the study focusing on the transport part, storage sites are assumed to be service providers in the presented model and incur only operational costs and emissions, without explicit investment costs. All storage sites have a maximum total and annual CO_2 storage capacity.

2.2. Modeling speed of pipeline construction

To model the limited speed and difficulty of constructing new transport technologies $j \in \mathcal{J}$, we include temporal diffusion limits based on the formulation by Leibowicz et al. (2016). Their formulation assumes that infrastructure investments are implicitly aggregated over a spatial region, as is common in energy system models. As the study focuses on the transport sector, the respective constraints are included only for the transport technologies $j \in \mathcal{J}$ and edges. The diffusion limit sets an upper bound on the capacity addition $\Delta S_{j,e,y}$ for each transport technology j on each edge $e \in \mathcal{E}$ per year y (Leibowicz et al., 2016):

$$\Delta S_{j,e,y} \leq \left((1+\vartheta_j)^{dy} - 1 \right) K_{j,e,y} + dy\, \zeta_j. \quad (8)$$

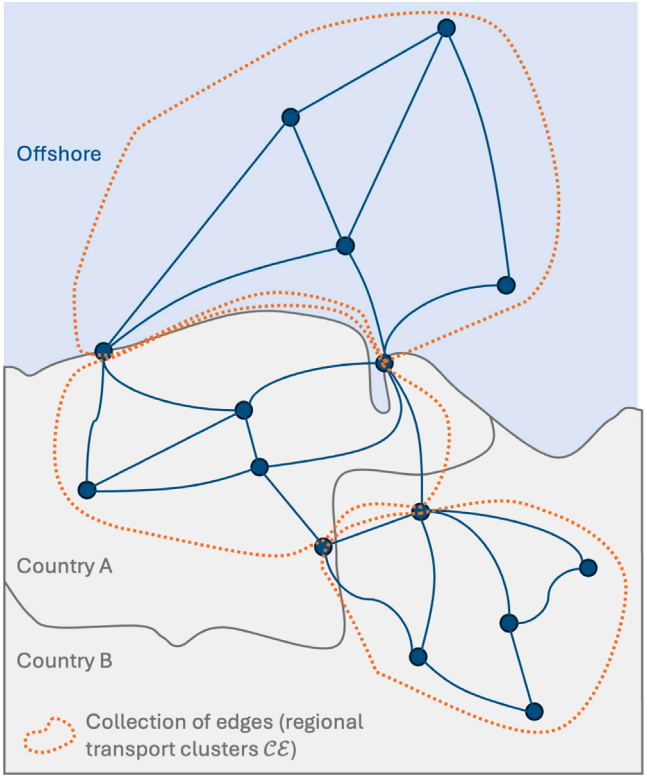


Fig. 1. Schematic CO_2 transportation network with nodes connected by individual edges. *Regional transport clusters* aggregate the edges of the network in order to apply the temporal diffusion limit. All other constraints apply to the individual edges and nodes instead of the transport clusters. The clusters only consider the edges, and the nodes are unaffected by the aggregation.

The limit for the capacity addition depends on two mechanisms: (1) building a knowledge stock $K_{j,e,y}$ over time and (2) entering a non-existent market, i.e., establishing a new industry without prior experience. The knowledge stock $K_{j,e,y}$ depends on previously built capacity additions $\Delta S_{j,e,y}$ of the same technology j and location e. As the knowledge stock grows with continued installations, the industry acquires growing installation capabilities over time. The increase of installation capabilities based on the knowledge stock depends on the *technology diffusion rate* ϑ_j, which behaves similarly to an annual growth rate. To account for the loss of knowledge over time, the capacity additions are multiplied by the knowledge depreciation rate δ (Leibowicz et al., 2016):

$$K_{j,e,y} = \sum_{\bar{y}=y_0}^{y-1} (1-\delta)^{dy(y-\bar{y})}\, \Delta S_{j,e,\bar{y}}. \tag{9}$$

The second mechanism for increasing installation capabilities is via pioneering projects, without an existing market. Pioneering projects refer to an industry's ability to install new technologies for the first time, i.e., without an existing knowledge stock. We model such projects by assuming a *pioneering capacity addition* ζ_j that can be installed in each year and investment period, even if $K_{j,e,y} = 0$ (Leibowicz et al., 2016). Therefore, two main parameters govern the potential speed of expanding infrastructure: The technology diffusion rate ϑ_j and the pioneering capacity addition ζ_j. We derive values for both parameters from currently announced pipeline projects in Europe (see Section 3).

We base the constraints on Leibowicz et al. (2016) but formulate the problem using *regional transport clusters*, which act as an aggregation layer of individual technology capacities (Fig. 1). Aggregating the individual connections allows modeling single edges while applying the diffusion limit to a cluster of transport connections. Energy system models commonly aggregate many technology installations into a single location, representing lumped production or transport capacity (Morbee et al., 2012; d'Amore et al., 2019). The regional transport clusters avoid formulating the diffusion limit for single transport connections, which would not reflect reality: On the one hand, single connections do not continuously expand the same technology to grow their local installation capability. As a result, the size of the pioneering capacity addition would entirely govern their capacity limit, and the diffusion rate would become irrelevant. On the other hand, one industry can serve the demand of a region or even a whole country rather than just a single location. Therefore, a limit on single edges does not accurately reflect reality. Temporal diffusion limits apply to expansion within regional transport clusters rather than at separate locations. The diffusion limit based on regional clusters can be applied to all types of technologies, whether they are node- or edge-based. Based on the aggregation, Eq. (8) is adjusted, and knowledge accumulation is accounted for over all edges e within the regional transport cluster $\mathcal{CE}$, and results in

$$\sum_{\bar{e}\in\mathcal{CE}} \Delta S_{j,\bar{e},y} \le \left((1+\vartheta_j)^{dy}-1\right) \sum_{\bar{e}\in\mathcal{CE}} K_{j,\bar{e},y} + dy\ \zeta_j^{cap}. \tag{10}$$

The diffusion limit in Eqs. (9) and (10) represents scale-up difficulties only with the capacity of the connection, i.e., how much CO_2 per year (Mt_{CO_2}/year) a pipeline can transport. In building pipelines, the challenge does not stem solely from transport capacity but also, to a substantial extent, from the distance: Longer transport distances require engagement with a larger number of communities and the acquisition of additional rights-of-way for the pipeline corridor. Additionally, the likelihood of crossing a state or country border increases, potentially introducing new regulatory hurdles or necessitating bilateral agreements. A purely capacity-dependent diffusion limit, suitable for technologies such as photovoltaics, may not adequately represent the challenges of scaling up pipelines. To address this limitation, the diffusion limit includes the distance by incorporating binary variables that represent pipeline construction.

For this purpose, we introduce two additional versions of the diffusion limit. Both versions take the capacity-based diffusion constraint in Eq. (10) as a basis and adapt the knowledge stock equation Eq. (9). One version uses the distance as the sole determinant of the diffusion limit, excluding the pipeline diameter from the constraint. The second version includes the product of capacity and distance, thereby accounting for both aspects of pipeline construction simultaneously. Consequently, pipeline expansion can be constrained in one of three ways: by capacity, based on annual flow (in Mt_{CO_2}/year, Eqs. (9) and (10)); by distance (in km, Eqs. (11) and (12)); and by their combination, the capacity-distance (in Mt_{CO_2} km/year, Eqs. (13) and (14)). All three approaches share the same fundamental diffusion parameters: the diffusion rate and the pioneering capacity addition term. The diffusion rate is dimensionless and therefore applicable to all formulations. The pioneering capacity addition term differs in magnitude and units to reflect the absolute size of initial installations.

$$\sum_{\bar{e}\in\mathcal{CE}} g_{j,\bar{e},y} \cdot d_{j,\bar{e}} \le \left((1+\vartheta_j)^{dy}-1\right) \sum_{\bar{e}\in\mathcal{CE}} K_{j,\bar{e},y}^{dist} + dy\zeta_j^{dist}, \tag{11}$$

$$\text{where}\quad K_{j,e,y}^{dist} = \sum_{\bar{y}=y_0}^{y-1} (1-\delta)^{dy(y-\bar{y})}\, g_{j,e,\bar{y}} \cdot d_{j,e}. \tag{12}$$

$$\sum_{\bar{e}\in\mathcal{CE}} d_{j,\bar{e}} \cdot \Delta S_{j,\bar{e},y} \le \left((1+\vartheta_j)^{dy}-1\right) \sum_{\bar{e}\in\mathcal{CE}} K_{j,\bar{e},y}^{cap-dist} + dy\zeta_j^{cap-dist}, \tag{13}$$

$$\text{where}\quad K_{j,e,y}^{cap-dist} = \sum_{\bar{y}=y_0}^{y-1} (1-\delta)^{dy(y-\bar{y})}\, d_{j,e} \cdot \Delta S_{j,e,\bar{y}}. \tag{14}$$

The additional constraints to limit deployment are all linear or mixed-integer constraints and do not require additional binary variables. Without deployment limits, capacities can be sized in each planning period y to meet demand perfectly or to reduce emissions

as required. The deployment limit introduces a temporal coupling between investment decisions, as the limits in each investment period depend on the installation decisions made in previous years. Linking investment decisions over time adds complexity and increases computational time.

3. Case study

3.1. Switzerland's needs for CO_2 transport

The CCTS infrastructure for Swiss industrial point sources is used to study the effect of the limited speed of pipeline construction on infrastructure development. Industrial CO_2 capture is an integral part of the Swiss decarbonization strategy to reach net-zero CO_2 emissions in 2050. The strategy includes a plan for the amount of CO_2 captured annually (BFE, 2021). However, without domestic geological CO_2 storage sites, Switzerland will depend on a transport network to export CO_2. As a landlocked country, offshore CO_2 storage depends on neighboring countries constructing sufficient pipeline capacity and permitting cross-border vehicle-based transport. Due to the comparatively well-developed plans for geological CO_2 storage and the need for export and long-distance CO_2 transport, Switzerland's industrial sector offers a well-suited setting for analyzing the limits of developing a transport network. The model optimizes infrastructure development from 2026 through 2050 and employs 3-year planning periods to circumvent computational constraints. Therefore, the transition pathway contains 9 planning periods, starting in 2026. The discount rate r for annualizing investment and discounting future costs is 6 %, based on the expected range for cost of capital for CO_2 transport (Sievert et al., 2025).

The point sources of CO_2 in Switzerland comprise mostly cement and waste incineration plants. Additionally, chemical plants and biomass-fired power plants are in operation. For the case study, we obtain the emission data for the point sources from the Swiss pollutant release and transfer registry 2023 (Swiss Federal Office for the Environment FOEN, 2024). The individual emitters are not differentiated regarding their willingness to participate in a CCTS infrastructure.

For all emitters, a generic post-combustion capture technology serves as a proxy for simplicity, as this study focuses on the transport infrastructure. Under the assumption of similar CO_2 purity specifications, likely requested by transport and storage providers (Northern Lights JV DA, 2024a), the choice of capture technology has a negligible impact on the transport infrastructure. Further, we assume high capture rates of 99 %, which are feasible and increase costs only slightly compared to lower capture rates of 90 % (Brandl et al., 2021; Pérez-Calvo and Mazzotti, 2022). Current projects often aim for capture rates around 90 % (Clean Air Task Force, 2024a). However, today companies offer a range of technologies that exceed the 95 % threshold (Barlow et al., 2025), suggesting that future projects may aim for higher capture rates. This trend is confirmed by recent studies on the role of CO_2 capture in the power sector, which quantify the benefits of higher capture rates (Homaei et al., 2026). Assuming a lower capture rate in the case study would require additional facilities to be connected to reach the same CO_2 sequestration target. Consequently, the demand for transport would increase, intensifying the effect of slow pipeline development on the expansion of the CO_2 network. The choice of 99 % is an optimistic but feasible assumption that does not fundamentally affect the transport design. Still, if current capture levels of 90 to 95 % remain the industry standard in the future, the need for timely transport deployment investigated in this study further increases.

The transport part of the CCTS chains relies on multiple transport modes (Section 2.1), namely trucks, trains, barges, and onshore pipelines for onshore and offshore pipelines and ships for offshore transport. The network structure, including the possible transport types between nodes, is defined before the optimization. Using the locations of emitters, harbors on the North Sea, and transfer locations along the river Rhine, the onshore pipeline connections are selected based on a Delaunay Triangulation between these nodes (Morbee et al., 2012; Becattini et al., 2022; Gabrielli et al., 2022). All onshore nodes lie on the edges of the Delaunay Triangulation, except barges, which are more restricted and can only navigate the Rhine downstream of Basel. The offshore edges for pipelines are determined via Delaunay Triangulation between North Sea harbors and storage sites. While long offshore pipelines are feasible, repressurization becomes challenging beyond 500 km (Holm et al., 2025). These long segments are therefore excluded from the candidate network. However, the remaining edges of the candidate network can still connect all nodes. Therefore, the modeling choice to exclude long edges is deemed to have minor effects. Ships can navigate from every harbor on the North Sea directly to the storage sites. The cost parameters for the transport modes (Table 1) are taken from Becattini et al. (2022) with an adjustment factor for offshore pipelines (d'Amore et al., 2021a). Following standard practice in literature, we assume transport costs to be constant over time (Morbee et al., 2012; Becattini et al., 2022).

CO_2 storage is based on publicly announced storage sites in the North Sea, the area primarily targeted for the Swiss decarbonization strategy (BFE, 2021). Storage sites in the North Sea are the most developed in Europe and also outnumber those in other areas, such as the Adriatic Sea, Croatia, and Bulgaria (Clean Air Task Force, 2024c). The national strategy for CO_2 sequestration until 2050 specifies an amount of CO_2 to be sequestered (BFE, 2021) and is enforced as an annual minimum CO_2 amount stored ($e_y^{CO_2-stored}$) across all storage sites in Eq. (6). Cumulatively, the national strategy aims to sequester 53 Mt_{CO_2} through 2050. The government's targets do not account for life-cycle emissions from CO_2 sequestration, which are, thus, excluded from the constraint (Eq. (6)).

3.2. Diffusion speed scenarios

The case study explores 4 scenarios to assess the consequences of diffusion limits on the network development and potential strategies to overcome deployment bottlenecks.

1. Unconstrained deployment: No diffusion limit applies to the pipelines, and all transport technologies are available.
2. Constrained pipeline deployment: A diffusion limit applies to pipeline construction. However, only pipelines are available for CO_2 transport, representing a commonly assumed case in which the entire CCTS industry relies on pipelines and does not consider or develop alternative transport modes.
3. Constrained pipeline deployment with multi-modal transport: The diffusion limit applies to pipelines, but alternative transport modes are available and can be installed. The diffusion limit does not apply to the alternative transport modes due to their availability and scalability. A complementary scenario investigating the consequences of diffusion limits for all transport modes is included in the Supplementary Information (Section A.6) and shows that the diffusion limit for non-pipeline transport modes has a negligible effect on the rollout of the CO_2 network (Section B.1).
4. Constrained pipeline deployment with domestic storage: In addition to the assumptions of scenario 2, a domestic storage site is assumed to be available in Switzerland starting in 2040. The storage linearly increases the injectivity of CO_2 until it reaches its maximum injection capacity of 3 Mt_{CO_2}/year in 2050 (BFE, 2021). In 2050, around 43 % of annual CO_2 sequestration demand can be fulfilled domestically.

Comparing the cost of the scenarios requires the same level of decarbonization. If achieved CO_2 sequestration falls short of the targeted amount, the gap is assumed to be compensated via direct air capture, a permanent and technology-driven CO_2 removal solution. The total cost for compensating the emissions depends on the year in which

Table 1
Transport cost parameters for the different transport modes. Some parameters scale per distance d and are specific for each edge. The size of a CO_2 transport connection is measured in *tons of CO_2 per year* (tpy). Details on transport cost calculations can be found in Section A.5 and in the ZEN-garden documentation (Reliability and Risk Engineering lab, 2025).

Transport mode	CAPEX		OPEX		Source
	Per capacity $\left(\frac{€}{\text{ktpy}}\right)$	Per distance $\left(\frac{€}{\text{km}}\right)$	Fixed $\left(\frac{€}{\text{ktpy}}\right)$	Variable $\left(\frac{€}{\text{kt}_{CO_2}\,\text{km}}\right)$	
Pipeline onshore	1000	100 000	0	8	Becattini et al. (2022)
Pipeline offshore	1710	171 000	0	13.68	d'Amore et al. (2021a) and Becattini et al. (2022)
Ship	$9 \cdot d$	0	0	60	Becattini et al. (2022)
Barge	$17 \cdot d$	0	0	60	Becattini et al. (2022)
Train	$8 \cdot d$	0	30 000	160	Becattini et al. (2022)
Truck	$4 \cdot d$	0	44 300	480	Becattini et al. (2022)

Table 2
Diffusion parameters used in scenarios 2–4 and the sensitivity analysis. Results for the capacity- and the distance-based diffusion limits are in the Supplementary Information.

Parameter		Scenarios 2–4	Sensitivity analysis	Unit
Technology diffusion rate		12	0–50	%
Pioneering addition of	capacity	n/a	0.02–3.9	Mt_{CO_2}/year
	distance	n/a	15–300	km
	capacity-distance	160	8–620	Mt_{CO_2} km/year
Knowledge depreciation		0	0	%

the emissions have to be compensated and the time-dependent cost of installing and operating direct air capture (Sievert et al., 2024).

Two main parameters determine the speed with which the pipeline network can grow: the pioneering capacity addition ζ and the technology diffusion rate ϑ (Eq. (13)). While scenarios 2–4 assume a standard value for the diffusion speed, a subsequent sensitivity analysis of both parameters shows their influence on the results. The diffusion speed parameters correspond to phases within the scale-up of a new industry. According to Kazlou et al. (2024), the number of announced projects and their success rate determine the diffusion speed in the earliest phase, called the formative phase. The stable growth rate controls the subsequent acceleration and growth phases (Kazlou et al., 2024). Although our model does not explicitly distinguish between the different phases, we can translate these terms into the two parameters determining the deployment speed: The pioneering capacity addition corresponds to the realized projects within the formative phase and their size. The realized projects need to consider both announcements of projects and their success rates. Whether and how many projects succeed can only be assessed relatively late before the expected start of operation. In 2025, no large CO_2 pipelines operate in Europe, and projects are only announced, based on which we can estimate the pioneering capacity addition. The size of currently announced new CO_2 pipeline projects in Europe (see Fig. 4 and Supplementary Information Table 4) is used as a reference (Clean Air Task Force, 2024b). Based on these projects, pioneering capacity addition values range from 55 to around 3700 Mt_{CO_2} km/year. Historical success rates in CCTS projects have been reported to range from 20 % (Abdulla et al., 2021) up to 50 % (Martin-Roberts et al., 2021). Based on historical success rates, we assume 40 % of projects to be realized and multiply this success rate by the median size (in capacity, distance, and capacity-distance) of announced pipeline projects (Table 2).

Similarly, the growth rate is difficult to estimate before stable growth actually materializes (Nemet et al., 2023), as projections for growth of other technologies have shown: Historical estimations underestimated the take-off speed of photovoltaics, but also overestimated the growth of nuclear power generation (Hoekstra, 2017; IAEA, 1975; World Nuclear Association, 2025). Even the deployment of CCTS systems has already been overestimated, with one report estimating large-scale facilities with a combined annual capture capacity of 1 Gt_{CO_2} operational by 2025 (IEA, 2009). Kazlou et al. (2024) estimate that the realistic growth of CCTS capacity is unlikely to exceed, and most likely falls well below 40 %. Nemet et al. (2023) report even lower values for the growth of historical technologies between 1.1 and 14.3 %. In scenarios 2–4, we assume growth rates of 12 %, aligning with historically observed diffusion rates (Nemet et al., 2023). We use a value from the upper end of the range due to the substantially higher estimations of necessary CCTS deployment by Kazlou et al. (2024). To address the uncertainty, we conduct a sensitivity analysis to investigate combinations of the diffusion rate and pioneering capacity addition parameters that make CCTS deployment too slow to meet climate targets. In scenarios 2–4, pipelines require a minimum flow of 60 % of their maximum capacity to avoid unrealistically large unused capacities (Wang et al., 2014).

The resulting mixed-integer linear program is solved with Gurobi 13.0 (Gurobi Optimization, LLC, 2026). We conclude the optimization with an optimality gap of 1 % or after 24 h, which results in optimality gaps of up to 1.9 %. The model size reaches 300'000 continuous and 8300 binary variables.

4. Results

4.1. Unconstrained deployment

The unconstrained deployment scenario serves as a benchmark for the constrained technology diffusion. In this scenario, pipeline capacities can be expanded without limitations based on previously built capacities. While we account for a construction time, this merely translates into a delay at the start of the optimization horizon.

In the unconstrained case, the optimal network layout (Fig. 2) utilizes a large backbone pipeline to connect Switzerland to storage locations in the North Sea. The cross-border backbone pipeline is installed early on and subsequently expands within Switzerland. Two branches of larger pipelines collect the CO_2 from smaller feeder pipelines that connect the individual emitters. By 2050, almost all considered emitters feed their CO_2 into the network comprising 843 km of onshore cross-border backbone pipelines, 1700 km of offshore pipelines, and an additional 945 km of domestic pipelines. The final layout and pipeline distances align closely with a previous official study for a Swiss pipeline layout based on exact corridors (Saipem, 2020), with the total length differing by less than 10 %.

The pipeline capacities built in the unconstrained case exactly meet the sequestration targets, as few constraints restrict the planning and construction of CCTS infrastructure. The transport network consists solely of pipelines, as the cost minimization exploits the economies of scale of pipelines and plans far enough ahead (Fig. 3a). The optimization model can meet the demand for CO_2 transportation by rapidly expanding pipeline infrastructure. The network expands from no existing pipelines to 915 km within 3 years, which may be unrealistic for CO_2 transport (Neele et al., 2011; d'Amore and Bezzo, 2017).

To avoid repeated pipeline expansions and benefit from low costs per tonne of CO_2 transported, pipelines are built to their final required size in the first time step. However, the resulting large transport capacities lead to unrealistically low pipeline utilization rates in the early

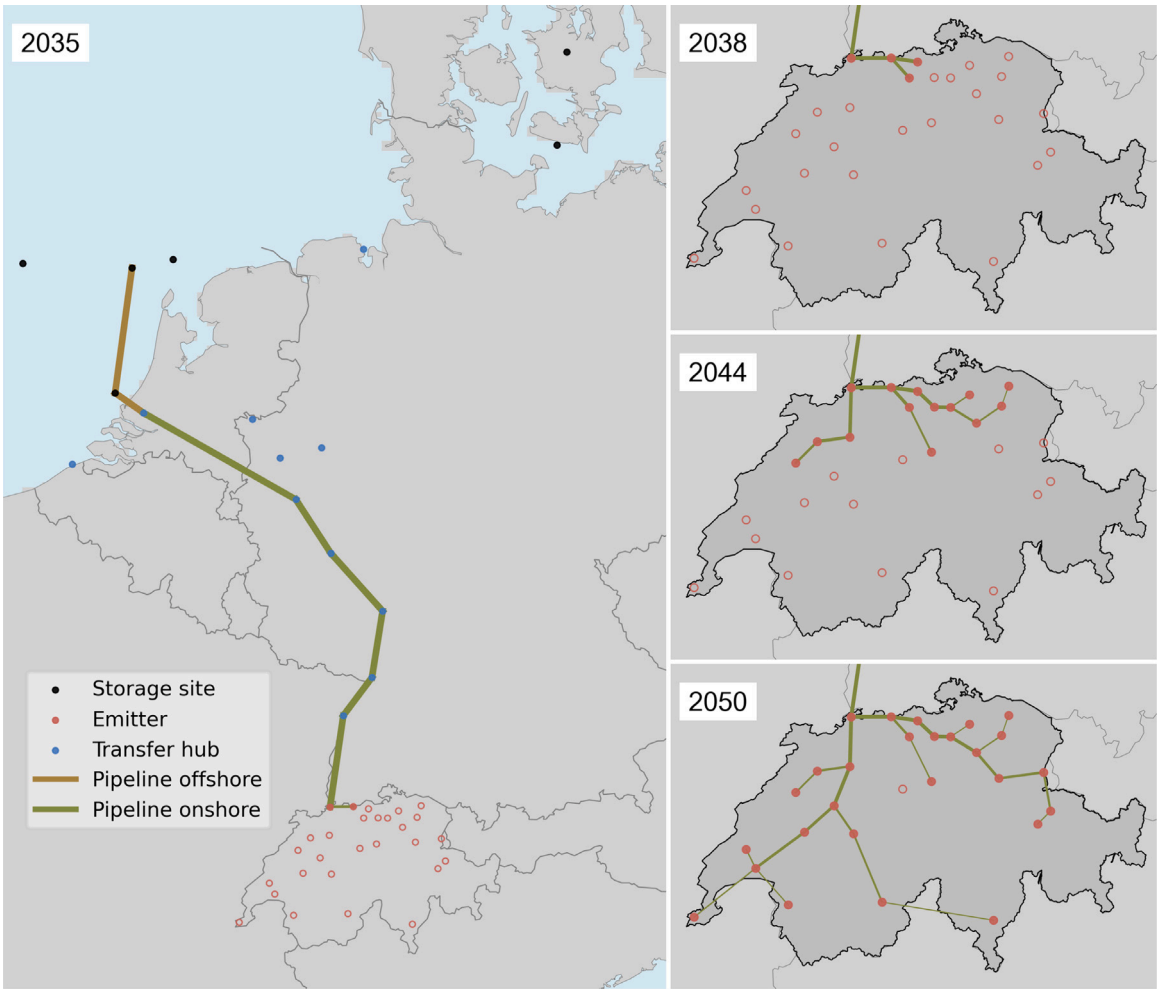


Fig. 2. Layout of transport infrastructure over time without deployment constraints.

years of the optimization horizon. In the first planning periods, from 2035 to 2038, on average, only 8 % of the installed flow capacity is utilized, as only a few emitters are capturing CO_2 at this early stage (Fig. 3a). For onshore pipelines, the effect is even more pronounced with 6 % utilization in the first and 18 % in the second planning period. Although no technical lower bound limits the flow through a pipeline, its operation becomes uneconomical if the ratio between design and operating capacity becomes too large: Wang et al. (2014) assess pipeline expansion strategies and conclude that oversizing is uneconomical compared to building multiple smaller pipelines if the utilization remains low for several years. Given the low initial flow rates and long time until all pipelines operate at their nameplate capacities (15 years in Fig. 3a), we conclude that oversizing is unrealistically high in the unconstrained scenario.

4.2. Constrained deployment

The unconstrained deployment scenario confirms previous studies, with pipelines emerging as the only required transport mode. Thus, the development of alternatives may seem unnecessary. In scenario 2, we consider pipelines as the only transport mode for CO_2 while trucks, trains, barges, and ships are excluded. However, expanding pipelines over time is now subject to the diffusion limit from Eqs. (13) and (14), which is based on installed capacity-distance to strike a balance between capacity and distance. The diffusion parameters are selected based on announced CO_2 pipeline projects and historical technology growth rates (Section 3).

The final layout of the network has features similar to those in the unconstrained case, with a backbone of large pipelines through Germany and branches collecting CO_2 from many regions of Switzerland (Supplementary Information Fig. 8). However, the transition pathway of expanding the network differs substantially (Fig. 3b). Under the deployment limit, pipeline capacities are expanded as fast as possible to store CO_2 early and to accumulate experience for larger capacity expansions later in the optimization. However, expansion remains lower than for the unconstrained scenario as the deployment limit becomes the defining constraint in the network expansion. Without alternatives to fall back on, the network cannot expand as fast as required. The constrained pipeline deployment fails to meet sequestration targets in several years between 2041 and 2050. Over these 9 years, the infrastructure development falls short of the goal by 4.8 Mt_{CO_2} cumulatively, which corresponds to 9 % of the cumulative sequestration target (BFE, 2021). In some years, the annual sequestration target is missed by 23 %. Therefore, infrastructure relying exclusively on pipelines may not develop as fast as required to meet policy goals.

Total pipeline deployment is highly sensitive to both parameters that govern its speed: pioneering capacity addition and technology diffusion rate (Fig. 4). With smaller deployment speeds, the gap between actually stored CO_2 and the policy target quickly approaches the overall sequestration target. For diffusion rates below 40 %, the success of the CCTS infrastructure depends on the scale of the initial installations and their failure rate. A pioneering capacity addition of 200 Mt_{CO_2} km/year for realized projects is required to ensure that the formative phase can effectively support the following acceleration and growth phase. For

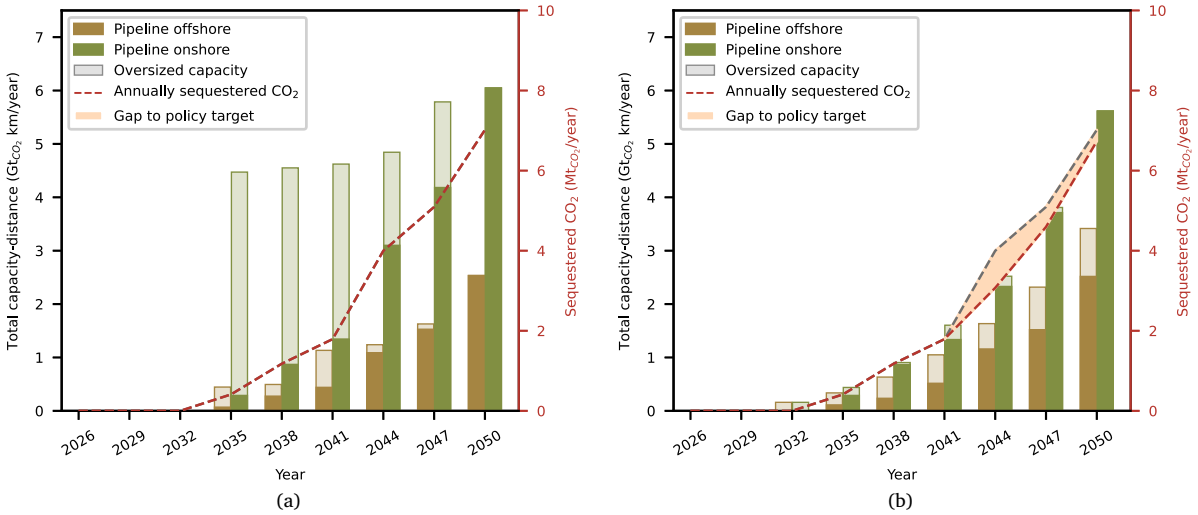


Fig. 3. Capacity expansion (light shaded bars) and utilized share (dark filled bars) of all transport modes over time. The right axes show the amount of CO_2 stored per year. Figure (a) shows the unconstrained case where all transport modes are allowed, and pipelines are not limited in their deployment speed. In Figure (b), the deployment limit for pipelines is active, and only pipelines can be built (scenario 2). Here, the cumulative sequestration target is missed by 1.7 Mt_{CO_2}.

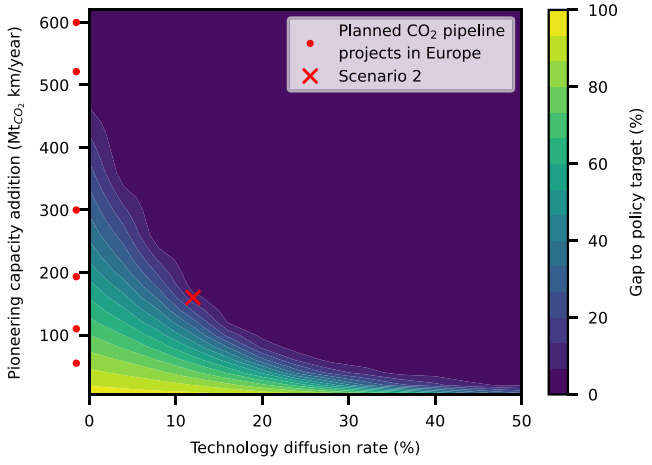


Fig. 4. Cumulative gap between achieved CO_2 sequestration and policy goal over the entire optimization horizon ($\sum_y E_y^{pen}$) depending on the unbounded construction and the technology diffusion rate. The gap quantifies the difference between the cumulative target (i.e., 53 Mt_{CO_2} through 2050) and the actual sequestration realized, normalized by the target. Announced European pipeline projects are indicated as red dots (Clean Air Task Force, 2024b). Two announced projects are outside the scale with pioneering capacity additions of 1100 and 3726 Mt_{CO_2} km/year respectively.

comparison, some of the currently announced CO_2 pipeline projects in Europe (red dots in Fig. 4) exceed this limit, although the success rate remains to be determined.

The range based on announced projects is, thus, an optimistic view of the development. As the largest projects have higher failure rates than smaller projects (Abdulla et al., 2021), the actual size of pioneering capacity additions will likely be considerably smaller. Therefore, the realizable value may lie within regions where scale-up cannot be achieved quickly enough. A similar picture emerges when basing the diffusion limit on either capacity or distance alone (see Supplementary Information Fig. 11).

4.3. Overcoming deployment limits

To overcome deployment limits of pipelines, two options are considered in this study: scenario 3, which uses non-pipeline transport modes to diversify the CO_2 transport sector and allows for bridging options until pipelines have developed, and scenario 4, based on domestic geological storage of CO_2, which is located closer to the emitters, thereby reducing the required transport distance. Results for both scenarios are presented below.

Scenario 3 allows all transport modes, including trucks, trains, and barges, while the deployment limit is active. As a result, barges and ships are included in the transport network in addition to pipelines and allow for meeting the sequestration targets (Fig. 3b and 5a). Installing barges eases the pressure on the onshore pipeline expansion, allowing the transport of CO_2 on the Rhine until sufficient pipeline capacity becomes available.

The network expansion can be divided into three stages. First, not much CO_2 needs to be transported during the early years of the CO_2 sequestration target, i.e., until 2041. The initial capacities of onshore pipelines suffice to transport all CO_2. For offshore transport, ships complement the pipelines in 2035, where the available amount of CO_2 is too small to operate the pipeline. However, after 2041, the demand increase in CO_2 sequestration accelerates. In these years, the pipeline construction cannot be scaled up quickly enough, and barges complement the onshore network to aid in scaling up. The barges transport 2 to 13 % of CO_2 (on a tonne-km basis) between 2044 and 2050. Towards 2050, more pipelines can be built, decreasing the need for barge transport. In 2050, the emerging pipeline network is strikingly similar to the unconstrained network of scenario 1. The total size of the transport system, including non-pipeline transport, is 8 % larger than the unconstrained scenario. Although the increase is small compared to the whole network, the alternative transport modes in scenario 3 enable reaching the climate targets compared to a system confined to pipelines as the ideal and only transport technology.

Instead of using non-pipeline transport, the required size of the transport system could be reduced if more local CO_2 storage sites are available. In most countries, storing CO_2 close to the emitters requires domestic onshore storage. However, in many European countries, onshore storage sites are undeveloped, only in the research phase, or even completely prohibited, although some countries are considering changes to legislation (Carbon Gap, 2025). In Switzerland, research into

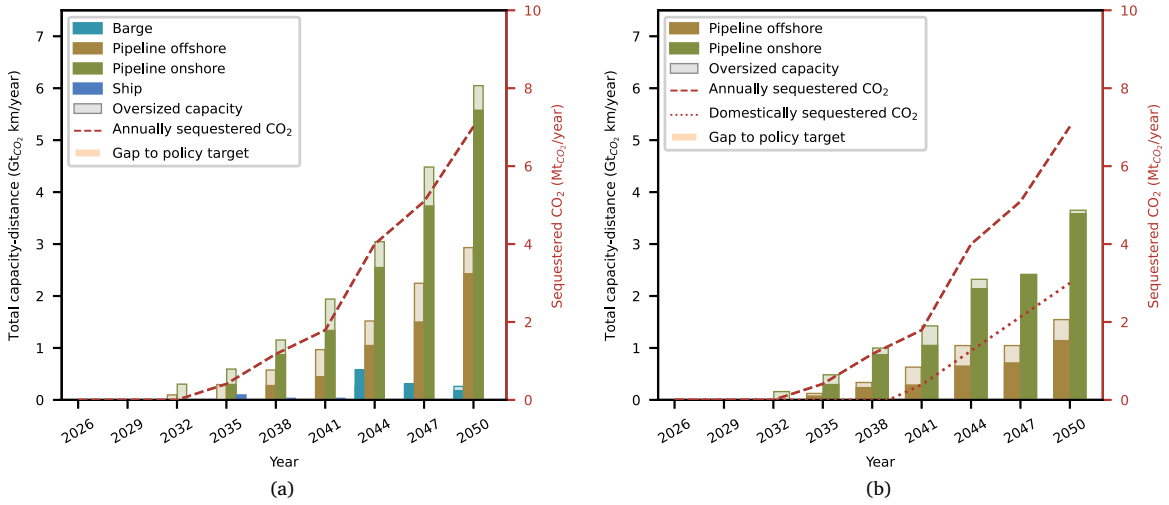


Fig. 5. Capacity expansion (light shaded bars) and utilized share (dark filled bars) of all transport modes over time with constrained deployment. The right axes show the amount of CO_2 stored annually. Figure (a) shows results from scenario 3, where multi-modal transport is allowed. In Figure (b), scenario 4 shows the benefits of having domestic storage space. Figure (b) includes the annual amount of CO_2 stored domestically. In both scenarios, the gap to the sequestration goal disappears, as both solutions overcome the deployment bottleneck.

domestic storage has started, and a pilot injection site will begin injecting CO_2 in 2026 (Becattini, 2025). A commercial-scale Swiss storage site is not expected before 2040 and is highly uncertain (BFE, 2021). When allowing for a domestic storage site in Switzerland while keeping the same deployment constraints, the gap between policy target and CCTS capability disappears (Fig. 5b). Domestic transport remains important for collecting CO_2 (Supplementary Information Fig. 10), but the connection to neighboring countries and the transit capacity do not need to be as large as in scenarios 1–3. Overall, the resulting transport infrastructure is around 40 % smaller than in scenario 1. The storage site opens in 2040 and, therefore, just in time to fill the gap to the policy target, which opens in the absence of a local storage site or alternative transport modes (Fig. 3b). However, if diffusion is slower than assumed in scenario 4, local storage capacity cannot prevent the policy target from being missed (Supplementary Information Fig. 12). In this case, the local storage needs to open earlier, or the domestic pipeline network expands not fast enough, creating a need for alternative transport modes despite local CO_2 storage. Although domestic storage may present a solution for many countries in their efforts to expand CCTS infrastructure, their success depends on many factors beyond the timely development of transport infrastructure (Lane et al., 2021). However, aspects connected to geological or societal uncertainty around permanent CO_2 storage are not analyzed in this study.

4.4. Multi-modal transport is cheaper than compensation by direct air capture

Alternative CO_2 transport modes may support ambitious transport infrastructure development. However, poor economies of scale compared to pipelines and high operational costs result in higher specific costs. In early-phase projects, which connect a single source with one sink, the costs have been reported to be 3–4 times higher than transporting the same amount in an advanced, shared infrastructure based on pipelines (Oeuvray et al., 2024; d'Amore et al., 2021a). Similarly, in the presented case study, using alternative transport modes increases cumulative costs by 5 % compared to the unconstrained scenario 1 (Fig. 6). While the cost of alternative transport modes is notably higher, their total capacity remains small compared to pipelines. The total cost of transport is still dwarfed by the cost for capture, conditioning, and storage (Fig. 6).

The constrained pipeline network in scenario 2 is smaller and therefore costs less than in scenario 1, but it fails to meet sequestration

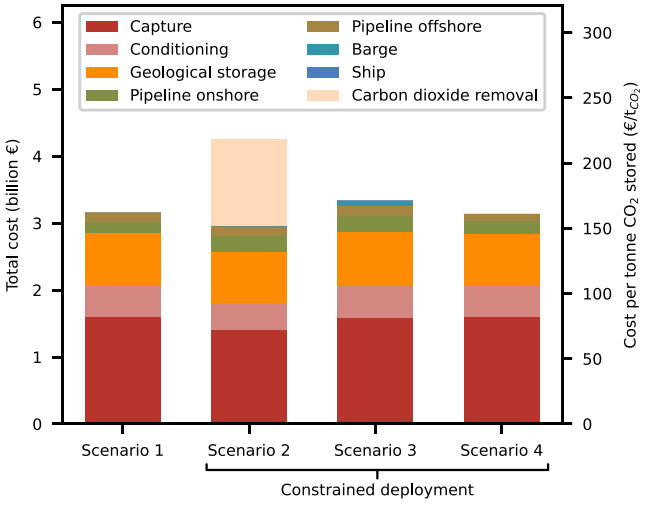


Fig. 6. Total cost and cost per tonne of CO_2 stored for all scenarios.

targets. With CO_2 removal, the overall cost of scenario 2 increases to 34 % above the unconstrained scenario 1 or to 27 % above that of the multi-modal case of scenario 3. Compared to direct air capture, the non-pipeline transport modes are a low-cost hedge against slow pipeline deployment. Given that these transport modes are flexible and can be relocated, the risk of long-term technology lock-in appears small. The costs of implementing a multi-modal transport system seem large in the short term, especially if used only for one emitter. However, when distributed across many emitters and the entire time horizon, the cost increase is marginal compared to the cost of compensating a CO_2 overshoot with direct air capture.

The cost of the CCTS value chain in scenario 4, including domestic storage, is similar to the cost of the unconstrained scenario 1. With capture, conditioning, and storage costs largely unaffected, the local storage influences only the transport infrastructure costs. However, the domestic storage can only sequester 20–43 % of the captured CO_2 in our case study. Thus, extensive exporting infrastructure investments are still required, reducing cost savings.

5. Discussion and conclusion

Transporting CO_2 has often been regarded as the trivial part in establishing CCTS value chains. The assumption hinges on the immediate availability of CO_2 pipelines. In Section 4, we show how a slow expansion of pipelines poses a threat to climate goals, which often hinge on the frictionless deployment of CCTS systems. Alternative transport modes are available to reduce the pressure on pipeline development and allow a faster scaling of the transport network in the long run. Thus, a more diversified suite of transport modes considered for CCTS chains lowers the risk of ending up with a network too small for the required CO_2 amount. Yet, the transport part is the least diversified part of the CCTS chain. Companies already offer a variety of capture technologies, and industrial and academic researchers actively work on developing new ones. Additionally, constructing capture units is typically faster and involves fewer stakeholders compared to pipelines. Similarly, several storage sites in the North Sea are either already operational or will be in the near future. Furthermore, several storage licenses are awarded each year in countries such as Norway, the UK, and Denmark. Therefore, with capture and storage being much further developed and not reliant on a single technology, a contingency plan is needed for slow pipeline deployment.

Non-pipeline transport modes are available, modular, and flexible, and could be deployed faster than large pipelines. Their implementation also requires infrastructure, such as transfer hubs, buffer storage units, and liquefaction units, but such installations are arguably less complex than a cross-border pipeline network. Planning a multi-modal transport network rather than a purely pipeline-based system hedges against delays and bottlenecks in pipeline deployment (Section 4.3). Such a network increases the planning certainty that is required for emitters to build capture plants.

Higher costs and operational complexity are the major disadvantages of using non-pipeline transport as a single emitter (Oeuvray et al., 2024). However, if the perspective is widened to the whole time horizon and the entire system of multiple emitters, the cost increase becomes small (Section 4.4). Given that the costs may fall within the overall transport cost uncertainty, a different approach seems possible: A large network operator could finance the infrastructure and redistribute the costs over time and across several emitters. In this case, the network operator can run non-pipeline transport modes and exploit their flexibility. The flexibility can be used to extend the CO_2 network and enable transport in new regions. Once a pipeline is available, these flexible transport means can be redeployed in the next region with newly developed capture projects. A large network operator could redistribute vehicles more easily than single emitters managing their own transport chains. Such redistribution reduces the individual cost of alternative transport modes and shares the risk across emitters.

In parallel to non-pipeline transport, local CO_2 storage sites reduce overall transport needs and can help overcome transport bottlenecks. Additionally, the siting of storage sites closer to emitters leads to lower costs and emissions for the whole CCTS system (Becattini et al., 2022). However, in Switzerland and many other European countries, no storage sites exist. Due to investment-intensive exploration and uncertain injection parameters, undeveloped local storage sites are unlikely to support CCTS scale-up in the near term. Therefore, non-pipeline transport modes offer a more feasible solution for developing a CCTS network in the near term than regional CO_2 storage.

The Swiss government's target, and thus the sequestration goal in our case study, does not include the life-cycle emissions associated with installing and operating the CCTS infrastructure. Therefore, the specific emissions of the transport modes do not affect whether the target is met. However, these emissions must be accounted for to ensure that the sequestered CO_2 is not offset by the carbon footprint of the CCTS system itself. Pipelines generally have lower specific greenhouse gas emissions than other transport modes (Nöhl et al., 2025). While CCTS chains based on multi-modal transport have a higher carbon footprint, they still store more CO_2 than they emit (Burger et al., 2024). In the context of this study, the CO_2 network emits more greenhouse gases per ton of CO_2 stored when using multi-modal transport. However, the cumulative life-cycle emissions from non-pipeline transport modes are small (200 kt_{CO_2}), corresponding to less than 5 % of the amount of CO_2 that remains uncaptured if relying only on pipelines (4.8 Mt_{CO_2}). Ultimately, to incentivize a network that is both timely and efficient, future policy targets should incorporate life-cycle emissions.

The optimization model shows how non-pipeline transport modes can overcome the potential deployment limits of pipelines (Section 4.3). However, other hedging strategies to address slow pipeline deployment are not included in the optimization. Industrial CO_2 emitters can choose other decarbonization options, although many industries have little or no alternatives besides CCTS (Davis et al., 2018). Another option would be to move production abroad to reduce domestic emissions and costs due to the ETS system. Policies including the European Union's Carbon Border Adjustment Mechanism target such "carbon leakage" (Grubb et al., 2022). However, if domestic solutions for decarbonization, like CCTS, fail to scale up, companies may still move production where conditions are more favorable for decarbonization. In such a scenario, the decision to move production abroad is caused by the industry's inability to decarbonize, rather than the financial pressure from ETS systems. Consequently, policymakers should not only focus on financial penalties for emitting industries but also on creating a favorable environment for scaling up decarbonization solutions, such as CCTS.

The effect of a favorable environment for scaling up is shown in the sensitivity analysis (Fig. 4). The diffusion parameters significantly influence whether the CO_2 network scales up successfully and achieves the climate targets. If diffusion parameters are high (e.g., $\vartheta > 20$ % and $\zeta > 200$ Mt_{CO_2} km/year), pipeline deployment is not the bottleneck for network scale-up. In fact, more than one solution to the scale-up problem is available for policymakers to consider: improving the diffusion speed through targeted regulation, supporting early projects to enable larger pioneering installations, diversifying transport modes, or exploring domestic CO_2 storage space. These many options offer opportunities to create favorable conditions for expanding the CCTS infrastructure, while balancing other aspects such as social acceptance. The presented framework for diffusion parameters provides a systematic tool for quantifying trade-offs among these options.

Our study introduces temporal diffusion limits of CO_2 pipelines in a transition pathway optimization. A case study of Swiss CO_2 sequestration goals within a European CCTS network demonstrates the consequences of slow pipeline deployment on climate targets. To enable future improvements of the method and applications besides our case study, the model and data are shared with open access (see Data Availability Statement). The results indicate that constrained pipeline deployment poses a significant risk for CO_2 sequestration goals. To mitigate this risk, non-pipeline transport modes offer flexible transport capacity that can be deployed quickly and can hedge against slow pipeline scale-up. Small additional transport capacities suffice to overcome pipeline buildout limits and increase total costs by only 5 % compared to an unconstrained scenario. Similarly, local CO_2 storage can reduce the risk of a slow pipeline network development, although its establishment is challenging due to long lead times and high geological uncertainty.

As no large-scale CO_2 pipelines are currently operational in Europe, deployment speed parameters remain uncertain until the construction of the first projects commences. Developing non-pipeline transport modes now hedges against the risk of slow pipeline deployment in the future.

CRediT authorship contribution statement

Johannes Burger: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **David Yang Shu:** Writing – review & editing, Supervision. **André**

Bardow: Writing – review & editing, Supervision, Project administration, Funding acquisition. **Giovanni Sansavini:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of generative AI in the manuscript preparation process

The authors acknowledge the use of AI for grammatical editing (Grammarly) and for code development (Gemini v3). After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: All authors report financial support was provided by the European Union's Horizon 2020 research and innovation programme under grant agreement No. 101022487 (ACCSESS project). André Bardow has served on review committees for research and development at ExxonMobil and TotalEnergies, oil and gas companies that are also active in carbon capture, transport, utilization, and storage. André Bardow has ownership interests in firms that render services to industry, some of which may work on carbon dioxide capture, transport, utilization, and storage. The remaining authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijggc.2026.104698.

Data availability

The complete input data for this article are available at https://github.com/ZEN-universe/ZEN-models/tree/Beyond_pipelines_2026. They contain all scenarios and sensitivity analyses. The optimization model used for computing the results, a modified version of ZEN-garden, is available at https://github.com/johburger/ZEN-garden/tree/Beyond_pipelines_2026.

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